Homework 2

This work must be handed on Sunday March 8th, 2026. You will not be allowed to hand in your report later than this. The answers to the worksheet are to be written neatly and clearly. Simply answer the questions one-by-one, in a neat and ordered manner. This homework can be added as an Appendix to the report, and referred to as appropriate.

Exercise 1: A linear time-invariant system is described by the difference equation

$$y[n] = \sum_{k=0}^{5} k x[n-k]$$

The input to this system *is unit step* signal, denoted by $u[n]$, i.e.,

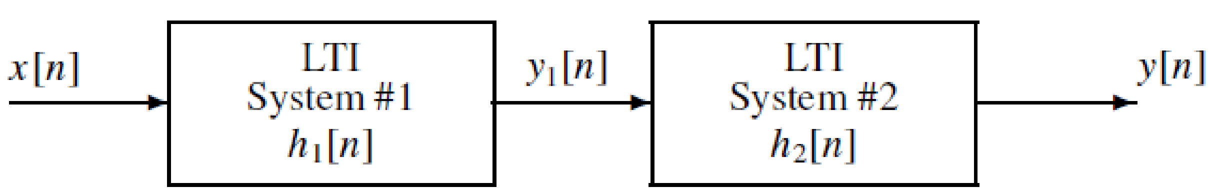
$$x[n] = u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

a) Determine the filter coefficients $\{b_k\}$ of this FIR filter.

b) Find the impulse response, $h[n]$, for this FIR filter. The impulse response is a discrete-time signal, so make a stem plot of $h[n]$ versus n.

c) Find the output $y[n]$ when the input is $u[n]$, and make a plot of $y[n]$ versus n.

Exercise 2: The following diagram depicts a *cascade connection* of two linear time-invariant systems, i.e. the output of the first system is the input to the second system, and the overall output is the output of the second system.



Suppose that System #1 is a seven-point running average filter,

$$h_1[n] = \begin{cases} 0 & n < 0 \\ 1/7 & n = 0, 1, 2, 3, 4, 5, 6 \\ 0 & n > 6 \end{cases}$$

And System #2 is described by the difference equation

$$y_2[n] = y_1[n] - y_1[n-1]$$

We can find the impulse response of the overall system, by first finding the impulse response of the first system and then use that response as the input to the second system.

a) Determine the filter coefficients of System #1, and also for System #2.

b) When the input signal $x[n]$ is an impulse, $\delta[n]$, determine the signal $y_1[n]$ and make a plot.

c) Determine the impulse response of the overall cascade system, i.e. find $y[n]$ when $x[n] = \delta[n]$. Take advantage of the fact that you already know the output of System #1, $y_1[n]$, when its input is an impulse.

Exercise 3: A linear time-invariant discrete-time system is described by the difference equation

$$y[n] = x[n] - 2x[n-1] + 3x[n-2] - 4x[n-3] + 2x[n-4].$$

a) Draw a block diagram that represents this system in terms of unit-delay elements, coefficient multipliers, and adders

b) Determine the impulse response $h[n]$ for this system.

c) Use convolution to determine the output due to the input

$$x[n] = \delta[n] - \delta[n-1] + \delta[n-2] = \begin{cases} 1 & n = 0, 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

Plot the output sequence $y[n]$ for $-3 \le n \le 10$.

Exercise 4: Let $x[n]$ be the complex exponential

$$x[n] = 11e^{j(0.3\pi n + 0.5\pi)}$$

If we define a new signal $y[n]$ to be the output of the difference equation:

$$y[n] = 2x[n] + 4x[n-1] + 2x[n-2]$$

it is possible to express $y[n]$ in the form

$$y[n] = Ae^{j(\omega_0 n + \phi)}$$

Determine the numerical values of $A, \phi,$ and ω_0.

Exercise 5: Let $x[n]$ be the complex exponential

$$x[n] = 7e^{j(0.1\pi n + 0.25\pi)}$$

If we define a new signal $y[n]$ to be the output of the FIR filter:

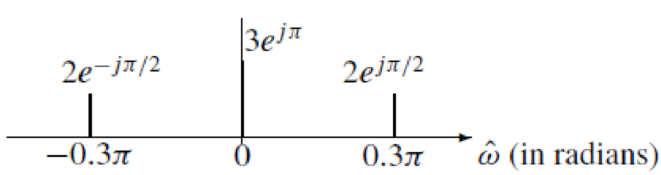
$$y[n] = x[n] + x[n-5]$$

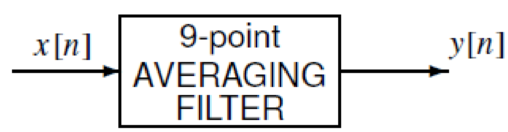
it is possible to express $y[n]$ in the form

$$y[n] = Ae^{j(\omega_0 n + \phi)}$$

Determine the numerical values of $A, \phi,$ and ω_0.

Exercise 6: A discrete-time signal $x[n]$ has the two-sided spectrum representation shown below.





a) Write an equation for $x[n]$. Make sure to express $x[n]$ as a real-valued signal.

b) Determine the formula for the output signal $y[n]$.

Exercise 7: The frequency response of this filter is

$$\mathcal{H}(\hat{\omega}) = \cos(\tfrac{1}{2}\hat{\omega})e^{-j\hat{\omega}}$$

If the input signal is

$$x[n] = 7 + 2\cos(0.5\pi n + \pi) \quad \text{for } -\infty < n < \infty.$$

determine a simple mathematical expression for the output signal $y[n]$.

Exercise 8: A linear time-invariant system is described by the difference equation

$$y[n] = 2x[n] + 4x[n-1] - 3x[n-2] + 3x[n-3] - 4x[n-4] - 2x[n-5]$$

a) Write a simple formula for the magnitude of the frequency response $|H(e^{j\hat{\omega}})|$. Take advantage of the odd symmetry of the filter coefficients (i.e., $b_0 = -b_5, b_1 = -b_4$, etc.) to express your answer in terms of real-valued functions only.

b) Derive a simple formula for the phase of the frequency response $\angle H(e^{j\hat{\omega}})$.

c) Determine the impulse response $h[n]$, and plot $h[n]$ as a function of n.

Exercise 9: Show that the DTFT function $X(e^{j\hat{\omega}})$ is always periodic in $\hat{\omega}$ with period 2π, that is,

$$X(e^{j(\hat{\omega}+2\pi)}) = X(e^{j\hat{\omega}}).$$

Exercise 10: Determine the DTFT of the following sum of two right-sided exponential signals

$$x[n] = (0.8)^n u[n] + 2(-0.5)^n u[n]$$